

# DRAMÉ BOURAMA

Mathematics and Computer Science Student (3<sup>rd</sup> year) | Aspiring Data Scientist & MLOps Engineer

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[Portfolio](#) | [Email](#) | [Github](#) | [LinkedIn](#) | [Kaggle](#)

## Featured Projects

### Real-Time Fraud Detection API & MLOps Architecture

[Code](#) | [Live API](#) | [Demo](#)

*FastAPI · Scikit-Learn · XGBoost · MLflow · Docker · GitHub Actions (CI/CD) · Streamlit*

- **Context:** Functional MLOps workflow for fraud detection and near-real-time transaction simulation.
- **Solution:** Logistic Regression vs. XGBoost comparison, custom Scikit-Learn transformers, MLflow tracking, Joblib serialization, Docker containerization, GitHub Actions CI, and a Streamlit dashboard connected to the API.
- **Evaluation:** ROC-AUC, classification reports, ROC curves, and feature importance.
- **Architecture:** FastAPI with Pydantic validation, prediction endpoint, JSON history of the 50 most recent transactions, and an HTTP simulator.
- **Business impact:** Quickly flags high-risk transactions, prioritizes human review, and provides a reproducible baseline for testing fraud-prevention strategies.

### E-Commerce MLOps API | Real-time Purchase Prediction (V2)

[Code](#) | [Live API](#) | [Demo](#)

*FastAPI · XGBoost · Docker · GitHub Actions (CI/CD) · Streamlit · Render*

- **Context:** V2 of the BigQuery project: evolved from a monolithic Streamlit app into a decoupled client-server architecture.
- **Solution:** HTTP service with Pydantic validation, pandas-based preprocessing, Docker, Pytest tests, GitHub Actions CI, and a Streamlit frontend consuming the API.
- **Features:** Status, health-check, and prediction endpoints, with validation of numeric inputs and supported countries.
- **Business impact:** Decouples the interface from inference, makes the model easier to reuse, and simplifies tested, deployed updates.

### Purchase Analysis & Prediction | GA4 + BigQuery (V1)

[Code](#) | [Demo](#)

*BigQuery SQL · XGBoost · Streamlit*

- **Problem:** Analysis of GA4 behavioral data to identify predictive purchase signals.
- **Solution:** SQL querying of nested BigQuery structures, feature engineering on 79,421 users, EDA, and XGBoost training.
- **Evaluation:** Precision, recall, and F1-score for the buyer class, and ROC curve, in a class-imbalance setting.
- **Business impact:** Surfaces behaviors linked to conversion, sheds light on drop-off points, and helps target the segments most likely to purchase.

## Technical Skills

- **Languages:** Python, SQL (BigQuery, PostgreSQL)
- **MLOps & DevOps:** FastAPI, Docker & Docker Compose, GitHub Actions (CI), Pytest, MLflow, Pydantic, Render Cloud, Swagger
- **Data Science & ML:** Pandas, NumPy, Scikit-Learn, XGBoost, Exploratory Data Analysis (EDA), feature engineering, Seaborn/Matplotlib
- **Tools:** Streamlit, VS Code, Git & GitHub

## Education

### Bachelor's Degree in Mathematics and Computer Science

2024 – Present

Faculty of Sciences of Bizerte | Tunisia

3<sup>rd</sup> year

- **Core coursework:** Linear Algebra, Big Data, Machine Learning, Databases, Probability and Statistics.

### Scientific Baccalaureate: Exact Sciences (Honors)

2023 – 2024

Specialization in mathematics and physics

## Certifications

- **Certifications:** Advanced SQL (Kaggle) | Data Manipulation with pandas & Intermediate Python (DataCamp) | Intermediate Machine Learning & Feature Engineering (Kaggle) | Seaborn & Matplotlib | Monitoring ML Concept (DataCamp)